VAMSI MIKKILI | M: +91 8688581103

E: mikkilivamsi3@gmail.com | Location: Chennai, India

Motivated Software Engineer with 3.8+ years (nearly 4) of total experience specializing in real-time embedded system architecture, Modern C++, Qt/QML, and Qt WebAssembly (Wasm) for mission-critical radar and defence applications. Eager to leverage technical expertise, analytical thinking, and Agile methodology to build high-performance software solutions.

PROFESSIONAL EXPERIENCE:

- Embedded Software Engineer with **3.8+ years** of proven experience in designing and developing high-performance, real-time embedded systems, specializing in **Modern C++**, **Qt/QML**, and **Qt WebAssembly (Wasm)** in **Linux/Windows** environments.
- Expertise in developing robust and scalable embedded applications, with a strong command over **Object-Oriented Programming (OOP)**, **Multi-Threading**, **Smart Pointers**, **Signal-Slot** event-driven architectures, and **Exception safety** in production-critical systems.
- Knowledge and hands-on experience with the Qt framework, including **Qt Quick**, **QML**, **Qt WebAssembly (Wasm)**, and **QAbstractListModel** for dynamic data visualization.
- Proficient in designing and implementing responsive GUI applications using Qt/QML, with seamless integration between QML frontends and **C++** backends.
- Experience in building modular and reusable UI components using **custom QML plugins**, contributing to clean architecture and faster development cycles.
- Extensive experience integrating hardware communication protocols including **RS-422/232**, **SPI**, **I2C**, **UART**, **UDP/TCP** sockets.
- Hands-on experience debugging applications using **QML Profiler**, **Valgrind**, and **GDB** for memory leak prevention and performance optimization.
- Skilled in **Linux**-based embedded system development, including work with custom **Yocto/BitBake** builds, cross-compilation, hardware register memory mapping (mmap), and hardware **Board bring-up**.
- Deep understanding of build systems such as **CMake**, **Make**, and **BitBake**, ensuring consistent and automated builds across development environments.
- Active contributor to **Agile** development workflows, with hands-on experience in **Jira** for sprint planning and issue tracking, **Git** for version control, and **CI/CD** pipelines for reliable deployments.
- Strong team player and mentor, regularly participating in code reviews, maintaining clean coding standards, and guiding junior engineers on best practices and code quality.

CORE COMPETENCIES:

- **C++**, **Qt/QML**, **Qt WebAssembly** | Yocto Project | UI/UX Architecture | Agile / Scrum | Git

EDUCATIONAL QUALIFICATION:

- Bachelor of Technology in Computer Science and Engineering — JNT University Kakinada, India.

SKILL SET:

Category	Technology & Tools
Programming & Scripting	C++ (C++11/14/17), Python, QML, JavaScript
Frameworks & Libraries	Qt 6 / Qt 5 (Qt Core, Qt Quick, QML), Qt WebAssembly (Wasm)
Hardware Protocols	RS-422, RS-232, SPI, I2C, UART, UDP, TCP/IP
Embedded & Build Systems	Yocto Project, Poky, BitBake Build System, Make, CMake, GCC / MinGW
Operating Systems	Windows, Linux (Ubuntu)
Development Tools	Qt Creator, Qt Maintenance Tool, VS Code
Debugging & Analysis	QML Profiler, Valgrind, GDB, Qt Inspector
Version Control & PM	GitHub, Jira, Taiga

WORK EXPERIENCE:

Organization	Period	Position
JEANUVS PVT LTD	August 2025- Present	QT Design Engineer
AIRAA CORPORATES	March 2024 – July 2025	Associate Tech Specialist
SYSTEM AIDS	Dec 2022 – Jan 2024	Engineer Trainee

PROJECT SUMMARY:

Project 1 : Avalanche Detection Radar (LAB - DRDO-DGRE)

Description:

A real-time embedded radar monitoring and control system developed for **DRDO-DGRE** to detect and issue early warnings for avalanche threats in high-altitude environments. Features dynamic target tracking, GIS map integration, multi-threaded hardware telemetry bridging, video stream integration, and automated alert mechanisms.

Technology: C++17, Qt6/QML, Qt WebAssembly, Embedded Linux, TCP/IP Sockets, SQLite, Systemd

Roles and Responsibilities Handled:

- Designed and developed a responsive QML-based GUI HMI for real-time radar target tracking, GIS map visualization, and operational control.
- Built a multi-threaded C++ hardware daemon bridge (radar_bridge) to parse real-time DSP data frames over memory-mapped interfaces (mmap).
- Implemented robust memory handling for hardware buffers and optimized real-time data flow between DSP hardware and backend servers.
- Interfaced with FPGA register controls for radar power sequencing, operating bandwidth configuration (10–15 MHz), and frequency activation checks.
- Developed backend telemetry infrastructure and SQLite database integration for real-time data streaming, user authentication, and automated alert management.
- Configured RTSP camera stream feeds, dynamic hooter alarm configurations, and automated log export utilities for system diagnostic analysis.

Project 2 : Naval Phased Array Radar System (LAB – INS VALSURA)

Description:

A high-performance naval phased array radar HMI and control software designed for real-time target scanning, tracking, and tactical display visualization. Provides continuous 360° rotational radar sweeps, target detection, and multi-plot spectral analysis across CW, FMCW, and Pulsed radar modes.

Technology: Modern C++ (C++17), Qt 6.11 (Qt Quick / QML), CMake, MinGW

Roles and Responsibilities Handled:

- Architected a 30 FPS real-time radar scanning engine using high-precision timers for smooth 360° sweep rotation.
- Implemented custom QAbstractListModel in C++ to manage simultaneous real-time tracking of military naval targets with range, azimuth, velocity, and signal strength attributes.
- Developed GPU-accelerated Qt Quick UI components for live Polar Radar Scopes, FFT Doppler Spectrum plots, and Range Profile visualizations.
- Built ISAR (Inverse Synthetic Aperture Radar) image display and 2D Range-Doppler Heatmap components for enhanced target classification.
- Integrated dynamic parameter configuration popups for switching radar operating modes (FMCW, Pulsed, CW) and controlling signal thresholds.

Project 3 : RF On-Board Data Retrieval System (RF-OBDRS) (LAB – DRDO –NSTL)

Description:

A mission-critical data retrieval and testing system developed for DRDO-NSTL to handle telemetry recording, peripheral diagnostics, prelaunch parameter configuration, and radio interface testing for naval defense applications.

Technology: Modern C++, Qt6/QML, Dual RS-422/RS-232, UDP Sockets

Roles and Responsibilities Handled:

- Developed master controller GUI and backend engine complying with DRDO-NSTL Interface Control Document (ICD) binary protocol with CRC-16 validation across 20 message types.
- Developed DualSerialManager in C++ for simultaneous bi-directional RS-422 and RS-232 serial communication with packet counting and error monitoring.
- Built high-speed UdpManager for real-time telemetry packet streaming, raw hex frame logging, and direct data recording to storage.

Project 4 : Embedded Manufacturing Flashing Tool

Description:

An automation tool for flashing firmware and software to embedded hardware boards such as NXP i.MX and Raspberry Pi 4, supporting user-selected build files, real-time device detection, hardware mapping, dynamic progress tracking, success/failure status, and detailed logging.

Technology: Qt/QML, C++, Yocto, Poky, Bitbake

Roles and Responsibilities Handled:

- Designed and developed a cross-platform flashing tool using Qt/QML and C++, implementing command-based firmware flashing over serial communication with device availability checks.
- Integrated real-time USB monitoring to detect hardware connection and disconnection dynamically, improving responsiveness and user interaction.

- Built modular backend classes and reusable UI components following Model-View-Presenter (MVP) architecture to ensure maintainability and scalability.
- Automated the flashing process using QProcess, with features like progress bar, success/failure status, and dynamic UI updates for improved usability.

PROFESSIONAL SKILL :

- Strong analytical and problem-solving mindset.
- Quick adaptability to new tools and technologies.
- Effective time management and task prioritization.
- Focused on delivering clean, efficient, and maintainable code.
- Good team collaboration and communication in Agile environments.

PERSONAL INFO :

- Nationality: Indian
- Current Location: Chennai, India
- Business Languages: English